UG (NEP) 4th SEMESTER

Applied Computing - IoT Fundamentals

ACP422N (Minor)

No. of Questions: 09 Date of Examination: 28-06-2025

Time Allowed: 2:00 Hours Max. Marks: 75 Min. Marks: 27

NOTE: Attempt all questions from Sections A and B, and only two questions from Section C.

Section A: Short Answer Type Questions (8 × 1.5 = 12 Marks)

Answer all questions in a maximum of 20 words each.

i. Define Actuators and Sensors.

ii. Write any two applications of IoT.

iii. Explain APIs briefly.

iv. What is the importance of Gateway?

v. What do you mean by Electronic Product Code?

vi. Write any two security issues of IoT.

vii. What do you mean by Smart Grids?

viii. What do you mean by Ultrasonic sensors?

Section B: Medium Answer Type Questions (4 × 7.5 = 30 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. Explain Basic components of IoT.

OR

Describe the Physical design of IoT.

3. Discuss Communication Models of IoT.

OR

Explain IoT reference Model.

4. Describe Hardware platforms in detail.

OR

What are necessary steps towards a security platform in IoT.

5. Write Home, Medical and smart city Applications of IoT.

OR

Explain NFC, 6loWPAN and ESP8266 codes.

Section C: Long Answer Type Questions (2 × 16.5 = 33 Marks)

Answer only two questions in a maximum of 400 words each.

6. Discuss characteristics of Internet of Things (IoT) in detail.

7. Write a detailed note on IoT Architectures.
8. Discuss how Basic Sensors are implemented.
9. What do you mean by IoT Protocols? Explain ZigBee and LoraWAN.